# Ventilator PERLS

Modes

A/C (Assist Control)– Delivers breaths at a set rate and volume. Each inspiratory effort beyond the set RR delivers a fixed tidal volume.

SIMV (Synchronized Intermittent Mandatory Ventilation) – Delivers breaths at a set rate and volume. Each inspiratory effort beyond the set RR are unassisted.

CPAP – (Continuous Positive Pressure Ventilation)- constant pressure is maintained throughout the respiratory cycle, no additional respiratory support (must trigger own breaths), tidal volume not set.

| Setting | Meaning | Normal |
|---|---|---|
| VT | Tidal Volume | 8-10cc/kg |
| PEEP | Peak End Expiratory Pressure | Usually >5 |
| FiO2 | Fractional inspired Oxygen | 21-100% |
| I:E | Inspiratory to Expiratory Time | 1:2 |
| PiP | Peak Inspiratory Pressure (Total pressure needed to push a volume of gas into the lungs | <35 normal lungs; <40 injured lungs |
| PP | Plateau Pressure- (End Inspiratory Pressure) | Usually <30cmH2O |

Alarms settings:

High setting – Usually 10-15 cmH2O above PiP

Low setting – Usually 10-15 cmH2O below PiP

<u>High Pressure Alarm</u>

Bronchospasm
Asynchronous breathing
Endobronchial intubation
Pneuomothorax
Kinked ETT
Secretions

Machine malfunction

<u>Low Pressure Alarm</u>

Ventilator disconnect
Airleak

<u>Complications of ventilation</u>
VALI – Pressures/O2
VAP
Hypotension
Ventilator dependence

**Modes of Ventilation** (mdc)

1. Volume Cycled [AC or IMV]
Rate: set (pt may breath above, each breath = set Vt)
Trigger: flow or pressure
Limit: flow
Cycle: volume
Peak Pressure: variable
Vt: constant
Advantages:
- guaranteed tidal vol.
Disadvantages:
- insp flow rate may not meet pt's demands thus increasing WOB
- airway pressure is variable and can lead to barotraumas

2. Pressure Control [PC]
Rate: set (pt may breath above, each breath = set pressure delivered)
Trigger: flow or pressure
Limit: pressure
Cycle: time
Peak Pressure: constant
Vt: variable
Advantages:
- improved gas distribution with longer insp time
Disadvantages:
- Vt varies with compliance

3. Pressure Support [PS]
Rate: variable (apnea backup rate only set)
Trigger: flow or pressure
Limit: pressure
Cycle: flow
Peak Pressure: constant
Vt: variable
Advantage:
- more comfortable for spontaneously breathing pt
Disadvantage:
- may increase pt WOB
- requires spontaneous resp drive

4. Synchronized Intermitent Mandatory Ventilation [SIMV +/- PS]
Rate: number of mandatory breaths are set, pt may breath above = either a CPAP or PS breath)
Trigger: flow
Cycle: Volume and/or flow
Peak Pressure: variable
Vt: constant for mandatory & synchronized breaths, variable for spontaneous breaths
Overview:
- pt receives 3 different types of breaths on this mode a) mandatory, b) synchronized, c) spontaneous
Advantage:
- ensures a minimum minute ventilation but also allows pt to breath above
Disadvantage:
- patient synchrony

4. Pressure Regulated Volume Controlled [PRVC]
Rate: set (pt may breath above, each breath = set pressure delivered)
Trigger: flow or pressure
Limit: pressure (variable)
Cycle: time
Peak Pressure: variable (max = 5cmH2O below high pressure alarm)
Vt: variable
Overview:
- ventilator monitors each Vt and adjusts pressure for the next breath to achieve a Vt within a target range (min-max).
- still requires close monitoring, pts can run into trouble with this

5. Airway Pressure Release Ventilation (APRV)
Rate: set (pt may breath spontaneously in addition at either the
Trigger: time
Limit: pressure
Cycle: time
Peak pressure: constant
Vt: variable
Overview:
- used in ALI
Advantages
- lower Paw for a given tidal volume compared with volume-targeted modes, e.g., AC, SIMV
-spontaneous breathing possible throughout entire ventilatory cycle
- decreased sedation use
- near elimination of neuromuscular blockade use
- improves lung recruitment & oxygenation
- constant airway pressure facilitates alveolar recruitment
- enhances diffusion of gases
- allows alveolar units with slow time constants to fill, preventing over-distension of alveoli
- augments collateral ventilation
Potential Disadvantages
-volumes change with alteration in lung compliance and resistance
-limited research and clinical experience